

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

MORE POS TAGGING IDEAS

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

- Part-of-Speech Tags
- POS Tagging using spaCy
- Classic POS Tagger using HMM
- **More POS Tagging Ideas & Summary**

Example features

- Word
- Lowercased word
- Prefixes
- Suffixes
- Capitalization
- Word shapes

Build a classifier to estimate $P(\text{tag} \mid \text{word})$

- → can lead to 93.7% accuracy without context

Conditional Random Field (CRF)

High-level explanations

- Extends the hidden states from discrete tags to continuous vector representations
- Builds a classifier to predict the tag based on the vector representation
- Decodes the tag sequence by considering the joint transition-emission probability distribution

Expectation-Maximization (EM)

- Start with a uniform HMM
- Run EM
- Inspect results

Clustering based on contexts

- Distributional representation
- Contextualized representations from neural language models
- ...

Accuracies degrade outside of domain

- E.g., applies a tagger trained on news to tweets
- Up to 3x error rate
- Usually make the most errors on the things you care about in the domain (e.g., protein names)
- This requires users to train their own domain-specific taggers

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